BRAIN TUMOR CLASSIFICATION USING HYBRID INTELLIGENCE

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Introduction

Brain tumors require early and accurate detection for effective treatment, and MRI is widely used due to its high diagnostic quality. However, manual interpretation of MRI scans is slow and prone to human error. While CNNs provide strong automated analysis capability, designing an optimal architecture and tuning its hyperparameters remains challenging. To overcome this, the proposed system uses a hybrid optimization framework combining Particle Swarm Optimization (PSO) for continuous hyperparameter tuning and Genetic Algorithms (GA) for discrete architectural refinement. This integrated approach automates CNN optimization and aims to improve accuracy and reliability in brain tumor segmentation from MRI images.

Literature Review

Naqvi *et al.* [1] optimized CNN hyperparameters using PSO for brain tumor classification and achieved better performance than manual tuning, but the method was limited by fixed search ranges and lacked architectural flexibility.

Deepthi and Jyothi [2] used handcrafted texture features with machine-learning classifiers, which worked on small datasets but lacked robustness and generalization for diverse MRI images.

Xie and Yuille [3] applied Genetic Algorithms to automatically evolve CNN architectures, improving performance, but the method was computationally expensive and struggled with continuous hyperparameter tuning.

Molina *et al.* [4] used U-Net and V-Net architectures for segmentation with good accuracy, but the models required extensive manual adjustment and did not generalize well across datasets.

Shanthi and Sabeenian [5] proposed a fuzzy logic–based segmentation method that was simple and fast, but highly sensitive to noise and low-contrast tumors, resulting in inconsistent accuracy.

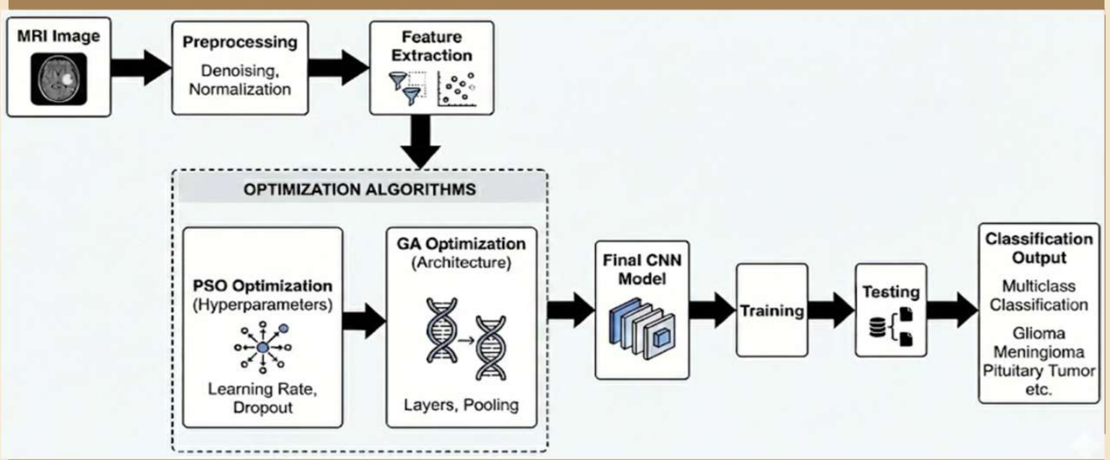
Project Objectives

1. To automatically detect and segment brain tumors from MRI images using a deep learning–based approach.
2. To optimize CNN hyperparameters and architecture using a hybrid PSO–GA framework.
3. To develop a highly accurate and robust model that outperforms existing brain tumor detection systems.

Proposed Methodology

The proposed methodology uses a **hybrid PSO–GA optimization framework** to automatically design and train a CNN for brain tumor classification. MRI images are first preprocessed and converted into feature maps using a CNN-based feature extractor. **Particle Swarm Optimization (PSO)** is then applied to tune continuous hyperparameters such as learning rate, dropout, and filter sizes. Next, **Genetic Algorithm (GA)** optimizes discrete architectural parameters like number of layers, activation functions, and pooling types. The best hyperparameters and architecture obtained from PSO and GA are combined to build the final optimized CNN model, which is trained using backpropagation and evaluated on test MRI scans for accurate tumor prediction.

Model Flowchart



Research Gaps

1. Existing PSO-based CNN optimization is limited by **fixed parameter ranges**, restricting exploration of better architectures.
2. PSO often **converges prematurely to local optima**, reducing model performance and generalization.
3. PSO cannot effectively optimize **discrete architectural parameters**, leading to limited CNN design flexibility.

Identification of Tools/ Technology / Algorithms

Python-3.8, Google Colab, PSO, GA, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib, Seaborn

References

[1] R. A. Naqvi *et al.*, "Automatic CNN architecture optimization for brain tumor classification using PSO," *J. Imaging*, vol. 11, no. 31, pp. 1–19, 2023.

[2] M. R. Deepthi and M. Jyothi, "Brain tumor detection using texture features and SVM," *IJERT*, vol. 9, no. 6, pp. 450–455, 2020.